15/12/18

Roll No. | | | | | | | | Paper Code: TCS 301 1-30-4.30

End Semester Examination DEC 2018

B.Tech (CSE) III Semester

Data Structure using 'C' language.

Time: Three Hours MM: 100

Note:

(i) This question paper contains five questions.
(ii) All questions are compulsory.
(iii) Instructions on how to attempt a question are mentioned against it.
(iv) Total marks assigned to each question are ***twenty.***

Q1. (Attempt any two questions of choice from a, b and c) (2X10=20 Marks)

a. *What do you mean by B tree and B+ tree? Construct a B tree of Order 3 with following keys : 16,3,12,15,1,2,4,5,8,9,*

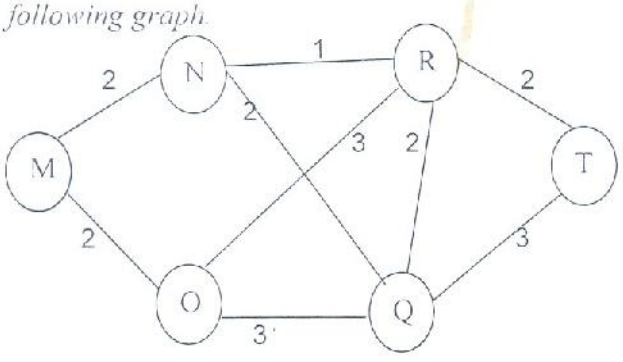
b. *What do mean by AVL tree and its balance factors. Draw an AVL tree with following Keys: L,V,A,X,P,Z,W,B,C,Y.*

c. *Explain hash collision with an example also explain any two hash collision resolving techniques.*

Q2. (Attempt any two questions of choice from a, b and c) (2X10=20 Marks)

a. Write a C function to sort an array using merge sort technique.

b. *Write an application of minimum spanning tree. Find minimum spanning tree of the following graph.*



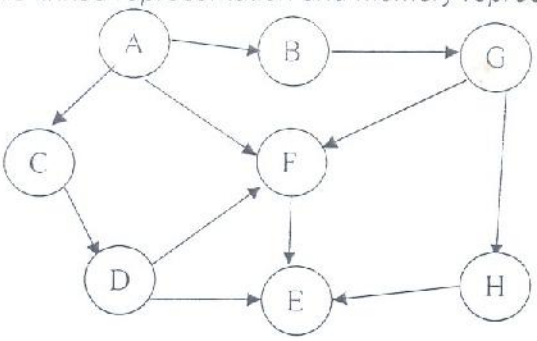
c. *Draw an expression tree from following infix expression:*
*Z= H%J/K^3 * 6-C*D^2+L*

Q3. (Attempt any two questions of choice from a, b and c) (2X10=20 Marks)

a. *Assume that you have a queue implemented with singly linked list concept, write a 'C' function to print that queue in reverse order that is from rear to front.*

b. *Explain index sequential file organization and relative file organization with examples.*

c. *Give linked representation and memory representation of following graph.*



Q4. (Attempt any two questions of choice from a, b and c) (2X10=20 Marks)

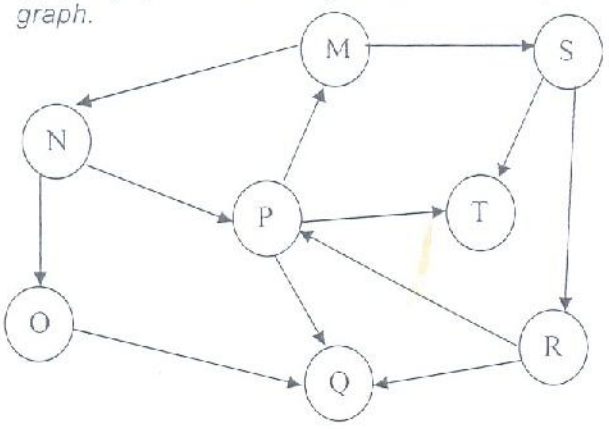
a. *Write applications of Huffman's algorithm. Using Huffman's algorithm encode following signal.*
b b d d b b a a c c c a c e c d c e c d c b b c b c b c

b. *Write a 'C' function to create a binary search tree also write a function to count total number of nodes from that binary search tree.*

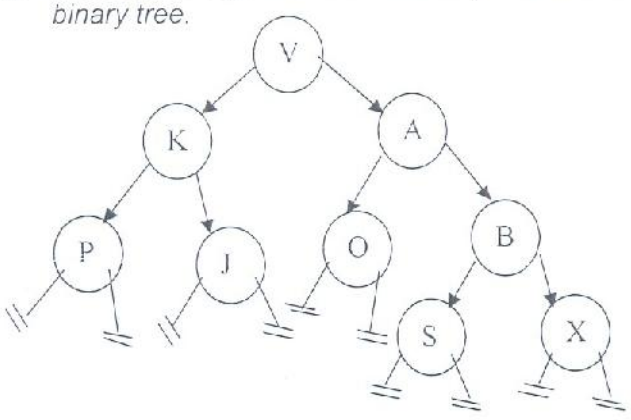
c. *Explain garbage collection and compaction with neat and clean diagrams.*

Q5. (Attempt any two questions of choice from a, b and c) (2X10=20 Marks)

a. *Apply a graph traversal algorithm to find all possible path from node M, on following graph.*



b. *Write advantage of threaded binary tree. Perform left and right threading on following binary tree.*



c. *Define hashing also explain various techniques of hashing.*